# 3.2 The Derivative as a Function

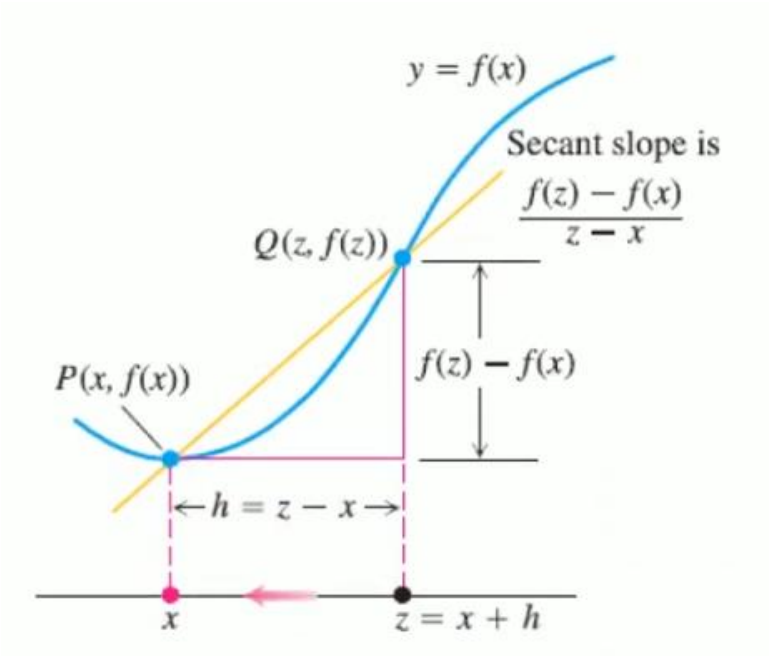
**DEFINITION** The **derivative** of the function $f(x)$ with respect to the variable $x$ is the function $f'$ whose value at $x$ is

$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h},$$

provided the limit exists.

**Alternative Formula for the Derivative**

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}.$$



## Calculating Derivatives from the Definition

**EXAMPLE 1** Differentiate $f(x) = \dfrac{x}{x-1}$.

**Solution** We use the definition of derivative, which requires us to calculate $f(x+h)$ and then subtract $f(x)$ to obtain the numerator in the difference quotient. We have

$$f(x) = \frac{x}{x-1} \quad \text{and} \quad f(x+h) = \frac{(x+h)}{(x+h)-1}, \text{ so}$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \qquad \text{Definition}$$

$$= \lim_{h \to 0} \frac{\dfrac{x+h}{x+h-1} - \dfrac{x}{x-1}}{h}$$

$$= \lim_{h \to 0} \frac{1}{h} \cdot \frac{(x+h)(x-1) - x(x+h-1)}{(x+h-1)(x-1)} \qquad \frac{a}{b} - \frac{c}{d} = \frac{ad-cb}{bd}$$

$$= \lim_{h \to 0} \frac{1}{h} \cdot \frac{-h}{(x+h-1)(x-1)} \qquad \text{Simplify.}$$

$$= \lim_{h \to 0} \frac{-1}{(x+h-1)(x-1)} = \frac{-1}{(x-1)^2}. \qquad \text{Cancel } h \neq 0.$$

**EXAMPLE 2**

**(a)** Find the derivative of $f(x) = \sqrt{x}$ for $x > 0$.

**(b)** Find the tangent line to the curve $y = \sqrt{x}$ at $x = 4$.

**Solution**

**(a)** We use the alternative formula to calculate $f'$:

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}$$

$$= \lim_{z \to x} \frac{\sqrt{z} - \sqrt{x}}{z - x}$$

$$= \lim_{z \to x} \frac{\sqrt{z} - \sqrt{x}}{(\sqrt{z} - \sqrt{x})(\sqrt{z} + \sqrt{x})}$$

$$= \lim_{z \to x} \frac{1}{\sqrt{z} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.$$

**(b)** The slope of the curve at $x = 4$ is

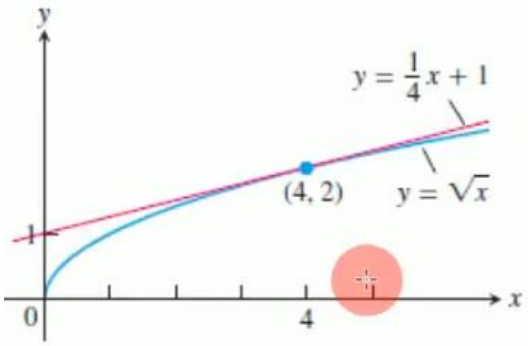
$$f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

The tangent is the line through the point (4, 2) with slope 1/4

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$$y = 2 + \frac{1}{4}(x - 4)$$

$$y = \frac{1}{4}x + 1.$$




## Notations

There are many ways to denote the derivative of a function $y = f(x)$, where the independent variable is $x$ and the dependent variable is $y$. Some common alternative notations for the derivative are

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = D(f)(x) = D_x f(x).$$

To indicate the value of a derivative at a specified number $x = a$, we use the notation

$$f'(a) = \left.\frac{dy}{dx}\right|_{x=a} = \left.\frac{df}{dx}\right|_{x=a} = \left.\frac{d}{dx} f(x)\right|_{x=a}.$$

For instance, in Example 2

$$f'(4) = \left.\frac{d}{dx}\sqrt{x}\right|_{x=4} = \left.\frac{1}{2\sqrt{x}}\right|_{x=4} = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

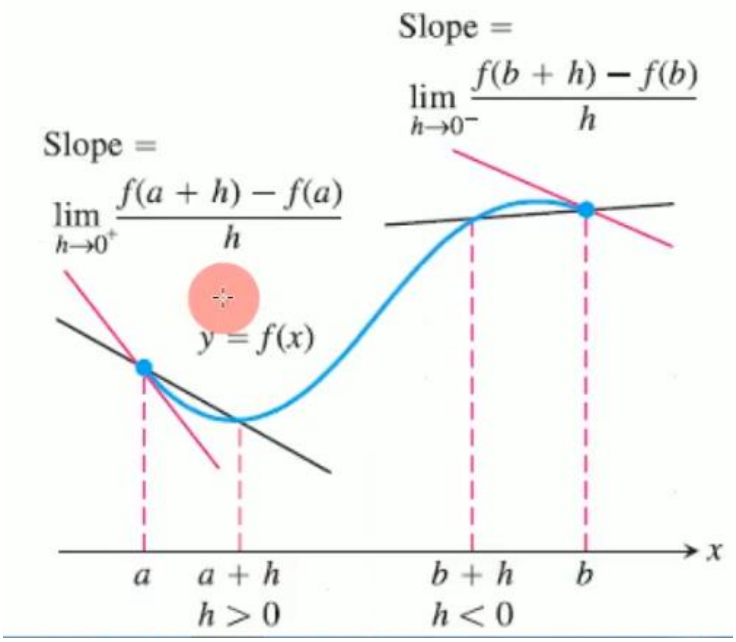
## Differentiable on an Interval; One-Sided Derivatives

A function $y = f(x)$ is **differentiable on an open interval** (finite or infinite) if it has a derivative at each point of the interval. It is **differentiable on a closed interval** $[a, b]$ if it is differentiable on the interior $(a, b)$ and if the limits

$$\lim_{h \to 0^+} \frac{f(a + h) - f(a)}{h} \qquad \textbf{Right-hand derivative at } a$$

$$\lim_{h \to 0^-} \frac{f(b + h) - f(b)}{h} \qquad \textbf{Left-hand derivative at } b$$

exist at the endpoints



## Remark

a function has a derivative at a point if and only if it has left-hand and right-hand derivatives there, and these one-sided derivatives are equal.

**EXAMPLE 4** Show that the function $y = |x|$ is differentiable on $(-\infty, 0)$ and $(0, \infty)$ but has no derivative at $x = 0$.

$$\text{Right-hand derivative of } |x| \text{ at zero} = \lim_{h \to 0^+} \frac{|0 + h| - |0|}{h} = \lim_{h \to 0^+} \frac{|h|}{h}$$

$$= \lim_{h \to 0^+} \frac{h}{h} \qquad |h| = h \text{ when } h > 0$$

$$= \lim_{h \to 0^+} 1 = 1$$

$$\text{Left-hand derivative of } |x| \text{ at zero} = \lim_{h \to 0^-} \frac{|0 + h| - |0|}{h} = \lim_{h \to 0^-} \frac{|h|}{h}$$

$$= \lim_{h \to 0^-} \frac{-h}{h} \qquad |h| = -h \text{ when } h < 0$$

$$= \lim_{h \to 0^-} -1 = -1. \quad \blacksquare$$

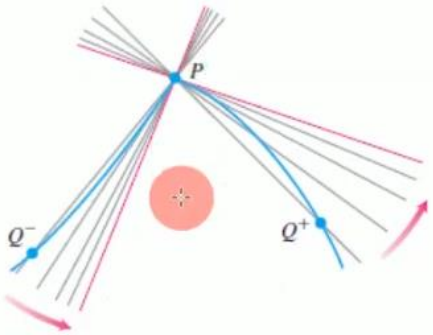
**EXAMPLE 5** In Example 2 we found that for $x > 0$,

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}.$$

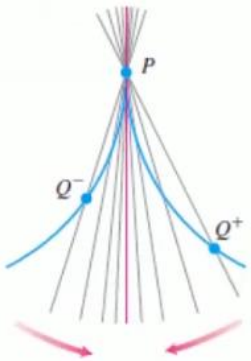
We apply the definition to examine if the derivative exists at $x = 0$:

$$\lim_{h \to 0^+} \frac{\sqrt{0 + h} - \sqrt{0}}{h} = \lim_{h \to 0^+} \frac{1}{\sqrt{h}} = \infty.$$

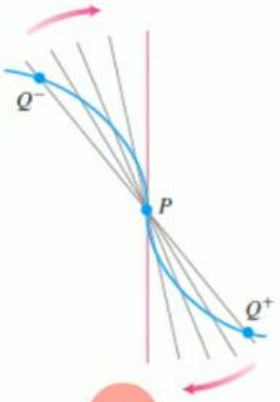
## When Does a Function *Not* Have a Derivative at a Point?



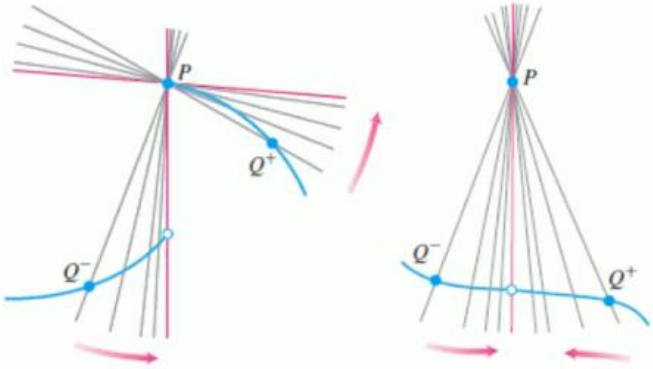

**1.** a *corner*, where the one-sided derivatives differ.




**2.** a *cusp*, where the slope of $PQ$ approaches $\infty$ from one side and $-\infty$ from the other.




**3.** a *vertical tangent*, where the slope of $PQ$ approaches $\infty$ from both sides or approaches $-\infty$ from both sides (here, $-\infty$).




**4.** a *discontinuity* (two examples shown).

## Differentiable Functions Are Continuous

A function is continuous at every point where it has a derivative.

**THEOREM 1—Differentiability Implies Continuity** If $f$ has a derivative at $x = c$, then $f$ is continuous at $x = c$.

## Remark

Theorem 1 says that if a function has a discontinuity at a point (for instance, a jump discontinuity), then it cannot be differentiable there. The greatest integer function $y = \lfloor x \rfloor$ fails to be differentiable at every integer $x = n$ (Example 4, Section 2.5).

The converse of Theorem 1 is false. A function need not have a derivative at a point where it is continuous, as we saw in Example 4.

# Exercises 3.2

In Exercises 17–18, differentiate the functions. Then find an equation of the tangent line at the indicated point on the graph of the function.

**17.** $y = f(x) = \dfrac{8}{\sqrt{x-2}}, \quad (x, y) = (6, 4)$

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{\dfrac{8}{\sqrt{x+h-2}} - \dfrac{8}{\sqrt{x-2}}}{h}$$

$$= \lim_{h\to 0}\frac{\dfrac{8\sqrt{x-2}-8\sqrt{x+h-2}}{\sqrt{x-2}\sqrt{x+h-2}}}{h} = 8\lim_{h\to 0}\frac{\sqrt{x-2}-\sqrt{x+h-2}}{h\left[\sqrt{x-2}\sqrt{x+h-2}\right]}$$

$$= 8\lim_{h\to 0}\frac{\sqrt{x-2}-\sqrt{x+h-2}}{h\left[\sqrt{x-2}\sqrt{x+h-2}\right]}\left(\frac{\sqrt{x-2}+\sqrt{x+h-2}}{\sqrt{x-2}+\sqrt{x+h-2}}\right)$$

$$= 8\lim_{h\to 0}\frac{x-2-(x+h-2)}{h\left[\sqrt{x-2}\sqrt{x+h-2}\right]\left(\sqrt{x-2}+\sqrt{x+h-2}\right)}$$

$$= 8\lim_{h\to 0}\frac{-h}{h\left[\sqrt{x-2}\sqrt{x+h-2}\right]\left(\sqrt{x-2}+\sqrt{x+h-2}\right)}$$

$$= 8\lim_{h\to 0}\frac{-1}{\left[\sqrt{x-2}\sqrt{x+h-2}\right]\left(\sqrt{x-2}+\sqrt{x+h-2}\right)}$$

$$\text{Slope} = f'(6) = \frac{-4}{(6-2)^{\frac{3}{2}}} = \frac{-4}{(4)^{\frac{3}{2}}} = -\frac{1}{2}$$

The equation of the tangent line is

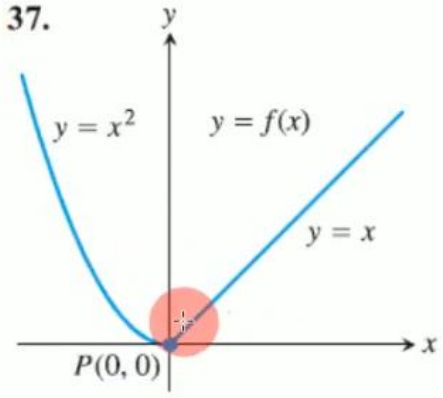
$$\frac{y-4}{x-6} = -\frac{1}{2}$$

$$y = -\frac{1}{2}(x-6)+4 \quad \Rightarrow y = -\frac{1}{2}x + 7$$

## One-Sided Derivatives

Compute the right-hand and left-hand derivatives as limits to show that the functions in Exercises 37–40 are not differentiable at the point $P$.

**37.**




$$\text{Right hand derivative} = \lim_{h\to 0^+}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0^+}\frac{f(h)-f(0)}{h} = \lim_{h\to 0^+}\frac{h-0}{h} = 1$$

$$\text{Left hand derivative} = \lim_{h\to 0^-}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0^-}\frac{f(h)-f(0)}{h} = \lim_{h\to 0^+}\frac{h^2-0}{h} = \lim_{h\to 0^+} h = 0$$

This implies that $f'(0)$ does not exist.